MICHAL BOREK

Machine Learning Engineer · Foundation Models, Multimodal AI & ML Systems
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SUMMARY

Machine Learning Engineer with 3+ years building and deploying multimodal AI and ML systems across medical imaging, foundation models, and distributed GPU infrastructure. Collaborates directly with CDC/NIOSH researchers and radiologists, combining rigorous evaluation, reliable AI, and production software from experimentation through deployment.

TECHNICAL SKILLS

Machine Learning & AI: Python, PyTorch, Hugging Face Transformers, MedGemma, LoRA/PEFT, Vision Transformers, ResNet, Multimodal AI, VLMs, Transfer Learning
Modeling & Evaluation: Zero-Shot Evaluation, ROC-AUC, Bootstrap Confidence Intervals, Calibration, Uncertainty-Aware Modeling, Ablation Studies, TensorBoard, Captum
ML Systems & Software: CUDA, H200 GPUs, PyTorch DDP, SLURM, ONNX, Docker, FastAPI, PostgreSQL, SQL, Git, pytest, GitHub Actions, CI/CD, AWS (S3, SageMaker, Lambda, API Gateway, ECR, CloudWatch)

EXPERIENCE

AI Research Consultant | Institute for Quantitative Health Science & Engineering, Michigan State University *Jun 2026 – Present*

- Collaborate directly with **CDC/NIOSH researchers and radiologists** on AI for pneumoconiosis and occupational lung disease; lead technical research reviews on foundation models, benchmarking, reliable AI, uncertainty, and deployment.
- Fine-tuned and benchmarked foundation models, improving held-out chest X-ray classification from **75% accuracy / 0.80 ROC-AUC to 85% / 0.96 ROC-AUC**; adapted MedGemma 1.5 4B with LoRA from **82% to 90%**.
- **Lead end-to-end development and beta deployment of an offline medical imaging AI application for radiologists**; engineered Python/PySide6 and ONNX software with **~1,860 automated tests**, Windows CI, deterministic inference validation, and automated installer packaging.
- Develop uncertainty-aware and fuzzy-labeling methods for ambiguous radiographic findings; mentor an undergraduate researcher stress-testing VLMs, Vision Transformers, ResNets, and related architectures.

Deep Learning Research Assistant | Institute for Quantitative Health Science & Engineering, MSU *Nov 2023 – May 2026*

- Built modular **PyTorch pipelines** for ordinal chest X-ray classification spanning preprocessing, training, calibration, evaluation, explainability, and reproducible experiment reporting.
- Scaled training with **PyTorch DDP on CUDA/SLURM**, reducing wall-clock training time from **~8 hours to 3 hours**; applied Captum analysis to investigate model failure modes.
- Developed monitored **AWS batch-inference workflows** with S3, SageMaker Batch Transform, Lambda, API Gateway, ECR, retry handling, and CloudWatch logging.

AI Engineer Intern, HPC / ML Tooling | Institute for Cyber-Enabled Research, MSU *Jun 2023 – May 2026*

- Built and deployed a **local AI agent** for SLURM job-script validation, combining deterministic checks, ShellCheck, local Ollama LLM review, Flask/CLI tooling, and Open OnDemand deployment.
- Developed an **ML resource-recommendation system** from historical cluster-job metadata to predict CPU, memory, GPU, and wall-time request bands before submission.

SELECTED PROJECT

Delta Dental Capstone – Project Lead — *Client-Sponsored Data Science, Spring 2026*

- Led a client-sponsored engagement with the **Delta Dental Data Science Team**, owning scope, technical direction, milestones, working sessions, and stakeholder presentations; recognized as the **Best Delta Dental Capstone Project**.
- Built a canonical analytics dataset by consolidating source tables into a stable schema and data dictionary; prototyped 3D feature mapping to improve interpretability and feature semantics.

EDUCATION

Michigan State University, College of Engineering *May 2026*
B.S. Computational Data Science, Minor in Mathematics · GPA: 3.80 / 4.00 · Honors College

PUBLICATION

Co-author, *"Ordinal Classification Framework for Multiclass Grading of Pneumoconiosis,"* SPIE Medical Imaging 2025, 13407:134072Q. DOI: 10.1117/12.3046353
Co-author, *"Pneumoconiosis screening and classification using deep learning models,"* Occupational & Environmental Medicine 2026;83(5):276-282. DOI: 10.1136/oemed-2026-111040